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Rewarding the Graph Behind the Chain: Topology-Aware Process Supervision for RL and Reasoning Distillation

 (/pdf?id=2JqC3rt4AV)

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 23 May 2026 (modified: 03 Jul 2026)  ACL ARR 2026 May Submission
 May, Senior Area Chairs, Area Chairs, Reviewers, Authors, Secondary Reviewers
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Keywords: process reward models, reinforcement learning, knowledge distillation, mathematical reasoning, chain-of-thought

Abstract:

Long chain-of-thought reasoning in LLMs is neither strictly correct-or-wrong nor purely linear. Outcome-only rewards overlook unjustified intermediate steps, while existing process reward models depend on step-level annotations and fail to capture the implicit conditional dependencies within a reasoning chain. We propose TopoPRM, a topology-aware implicit process reward model that extracts a dependency DAG from each reasoning trace and integrates conditional dependency and sequential continuity as complementary components of a multi-source reward signal. Building on topoprpm, we design TGSD, a post-training framework that combines supervised warm-start, GRPO optimization under a correctness-first hierarchical reward, and topology-guided on-policy distillation. Across multiple public benchmarks, our method improves average accuracy over standard GRPO at both 7B and 9B scales while generating up to 2% fewer tokens, with consistent gains on elementary and general-domain reasoning.

Paper Type: Long

Research Area: NLP and Symbolic Reasoning

Research Area Keywords: Machine Learning for NLP, NLP and Symbolic Reasoning

Contribution Types: Model analysis & interpretability, Approaches to low-compute settings (efficiency), Publicly available software and/or pre-trained models

Languages Studied: English

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: Yes

A2 Elaboration: This paper has a limitations section.

B Use Or Create Scientific Artifacts: Yes

B1 Cite Creators Of Artifacts: Yes

B1 Elaboration: §4 Experimental Part

B2 Discuss The License For Artifacts: No

B2 Elaboration: We use publicly available models under their original licenses but do not explicitly discuss license terms in the paper.

B3 Artifact Use Consistent With Intended Use: Yes

B3 Elaboration: §4.1
B4 Data Contains Personally Identifying Info Or Offensive Content: N/A
B5 Documentation Of Artifacts: Yes
B5 Elaboration: Appendix A.
B6 Statistics For Data: Yes
B6 Elaboration: §4.1 and Appendix A.
C Computational Experiments: Yes
C1 Model Size And Budget: Yes
C1 Elaboration: Appendix A
C2 Experimental Setup And Hyperparameters: Yes
C2 Elaboration: Appendix A
C3 Descriptive Statistics: Yes
C3 Elaboration: We report pass@1, pass@5, maj@5, Acc/kToken, and so on.
C4 Parameters For Packages: Yes
C4 Elaboration: §4.1 Setup and Appendix A
D Human Subjects Including Annotators: No
D1 Instructions Given To Participants: N/A
D2 Recruitment And Payment: N/A
D3 Data Consent: N/A
D4 Ethics Review Board Approval: N/A
E Ai Assistants In Research Or Writing: No
E1 Information About Use Of Ai Assistants: N/A
Author Submission Checklist: yes
EMNLP 2026 AI Reviewing Experiment: no
TLDR: 👁 We extract dependency graphs from reasoning traces and use their topology as a free process reward signal for both RL optimization and distillation.
Preprint: 👁 no
Preprint Status: 👁 We are considering releasing a non-anonymous preprint in the next two months (i.e., during the reviewing process).
Preferred Venue: 👁 EMNLP
Visa Needs: 👁 yes
Consent To Share Data: 👁 yes
Consent To Share Submission Details: 👁 On behalf of all authors, we agree to the terms above to share our submission details.
Association For Computational Linguistics - Blind Submission License Agreement: 👁 On behalf of all authors, I agree
Submission Number: 4012

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05 Jul 2026, 00:14 (modified: 12 Jul 2026, 23:21)

👁

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer HxUk

Revisions (/revisions?id=Qj0r8hDJNm)

Paper Summary:

This paper proposes TopoPRM, which models long reasoning traces as implicit dependency DAGs and combines global topology with local sequential continuity for process rewards. Based on this reward model, TGSD integrates supervised warm-start, GRPO, and topology-guided on-policy distillation, improving overall accuracy while reducing reasoning length across several benchmarks.

Summary Of Strengths:

- 1. Modeling non-local dependencies between reasoning steps offers a meaningful alternative to conventional PRMs that treat reasoning as a purely linear sequence.
- 2. The correctness-first hierarchical reward and ACE mechanism help prevent structurally coherent but incorrect solutions from receiving overly high rewards. The motivation is clear and supported by the ablation results.
- 3. The experiments cover diverse reasoning tasks and examine accuracy, token efficiency, training dynamics, component contributions, and cross-scale distillation. Overall, the results suggest improvements in both performance and reasoning efficiency.

Summary Of Weaknesses:

- 1. Since the dependency edges are identified through expression reuse, variable overlap, and LLM-based inference, it would be useful to compare them with human-annotated dependency graphs or report edge-level precision, recall, and agreement. This would make it clearer whether the topology reward reflects the actual logical dependencies in the reasoning traces.
- 2. The distillation experiments mainly compare TGSD with standard SFT distillation and off-policy KL regularization. Comparisons with stronger on-policy distillation, reasoning compression, or explicit length-control methods under comparable data and training budgets could provide a more convincing assessment.
- 3. Most experiments are conducted within the Qwen model family and its distilled variants. Evaluation on other model families, such as Llama, could better support the generality of the method. In addition, some competition benchmarks are relatively small, but results over multiple random seeds or significance tests are not reported.

Comments Suggestions And Typos:

N/A

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3.5

Excitement: 3.5

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Limitations And Societal Impact:

yes

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Official Comment by Authors 📅 14 Jul 2026, 13:11 (modified: 14 Jul 2026, 22:46)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer HxUk

📄 Revisions (/revisions?id=NsvpMgUelQ)

Comment:

We thank the reviewer for recognizing the motivation behind non-local dependency modeling, the correctness-first reward, and ACE. The concerns ask whether the gains come from topology itself rather than length control, one model family, or an unvalidated extractor. R1/R2/R3 below answer W1/W2/W3 in order.

W1. Edge-level validation against LLM and human references.

R1. This is the most direct test of whether the topology reward reflects reasoning dependencies rather than surface overlap, so we built an edge-validation set of 120 traces from GSM8K/MATH stratified by length (short/medium/long). For each trace we segment the solution into steps and enumerate ordered step pairs, then label whether step i is necessary support for step j under two independent references: a held-out Qwen3-32B annotator (scalable), and human annotation. Extractor precision/recall/F1 against each reference:

Table R1. Edge-level extractor precision/recall/F1 against the Qwen3-32B annotator and the human-adjudicated reference (120 stratified GSM8K/MATH traces).

Reference	Precision	Recall	F1
Qwen3-32B	0.48	0.59	0.53
Human annotation	0.58	0.71	0.64

Against the human reference the extractor reaches $F1=0.64$, higher than against the automatic annotator, indicating that the extracted edges align even better with human-judged necessary support once the automatic annotator's over-strict rejections are corrected. The edge-type breakdown is transparent about failure modes: expression/claim/order edges align well (precision 0.59–0.63), while variable-only overlap is the weak type (0.25) and accounts for most false positives — which we further tighten in the release with two conservative guards (variable edges require ≥ 2 shared variables; fallback order edges require local lexical overlap). To show the reference is reliable rather than a single model's idiosyncrasy, a second architecturally distinct annotator (Qwen2.5-32B) reaches **0.90 raw agreement and Cohen's kappa 0.73** (substantial agreement) with the first on the per-pair decision. Beyond this public-benchmark sample, TopoPRM originates from a deployed math-grading platform (Appendix F) where professional teachers annotate step-level support and correctness; those production traces provide a strict, human-authored dependency reference that motivated and continues to validate the extractor. We rescope the claim to **surface-evidenced support dependencies** rather than proof-level logical-dependency recovery, and will release the annotation guideline, edge-type confusion table, and representative DAG visualizations.

W2. Stronger distillation and length-control baselines.

R2. For distillation, TGSD improves over static SFT distillation [1] and off-policy KL [2] under the same 9B teacher and 4B student:

Table R2. 4B-student distillation under a shared 9B TopoPRM teacher: pass@1, token ratio vs. teacher, and structural retention (acyclicity/no-orphan/edge-keep on MATH-500).

4B student	GSM8K	MATH-500	Token ratio	Structural retention
Off-policy KL [2]	76.8	49.3	>1.0x	0.41x
Static SFT distillation [1]	79.4	58.2	0.64x	0.86x
TGSD (on-policy, topology-guided)	82.8	61.5	0.56x	0.93x

For length control, we ran a matched outcome+length GRPO baseline with the same length regularizer but no topology/continuity signal. On a controlled GSM8K subset, outcome-only, outcome+length, and full TopoPRM score 75.5, 76.5, and 77.0 pass@1. We treat this as an attribution check, not a headline:

length control helps but does not fully explain the TopoPRM gain, and TGSD's advantage comes from the topology-conditioned revision target rather than the distillation loss, so it composes with explicit length-control methods [3, 4] rather than competing with them. To isolate this directly, the revision adds a plain on-policy distillation/revision baseline that uses the same self-generated traces and budget but *without* the topology-conditioned revision prompt, so any remaining gap is attributable to the topology signal itself rather than to on-policy sampling.

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Response to Reviewer HxUk (2/2)

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by Authors (👤 Zhuoming Zhu (/profile?id=~Zhuoming_Zhu1), Xixuan Hao (/profile?id=~Xixuan_Hao2), Songxin Lei (/profile?id=~Songxin_Lei1), Yuxuan Liang (/profile?id=~Yuxuan_Liang1), +2 more (/group/edit?id=aclweb.org/ACL/ARR/2026/May/Submission4012/Authors))

📅 14 Jul 2026, 22:46

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer HxUk

Comment:

W3. Generality and statistics.

R3. The main experiments already span three distinct backbone families, not one, so the gains are not tied to a single model: Qwen2.5-7B (base), Qwen3.5-9B , and DeepSeek-R1-Distill-Qwen-7B (a distillation of R1 into a separate lineage), where TopoPRM raises the nine-benchmark average to 53.0, 58.6, and 58.3, respectively (Table 1). We deliberately run the matched RL comparisons on DeepSeek-R1-Distill-Qwen-7B because it is a trainable, non-saturated base. Already heavily instruction-tuned models such as Llama-3.1-8B-Instruct instead sit near the GSM8K ceiling (84.5 pass@5, Table 1; comparable strong instruction-tuned baselines such as Llama-3.3-70B-Instruct reach only 82.9 GSM8K F1 in the reranking study of [5]), leaving little accuracy headroom for any process reward and making them uninformative as an RL testbed on this axis. Our claim is a topology-aware process signal whose benefit shows on backbones with headroom, and it is evidenced across the three families above rather than on a single family. For statistics, we run three seeds of the matched DR1-7B comparison and report uncertainty:

Table R3. Matched DeepSeek-R1-Distill-Qwen-7B comparison over three seeds: GSM8K pass@1 (mean ± std) with Wilson 95% confidence intervals (n=200).

Reward	GSM8K pass@1	Wilson 95% CI (n=200)
Outcome-only GRPO	75.5 ± 0.4	[69.1, 80.9]
Full TopoPRM	76.0 ± 0.7	[69.6, 81.4]

The intervals overlap on this saturated subset, so we do not claim a significant single-benchmark win; the substantive accuracy gains appear on the nine-benchmark average (58.3 vs 55.1 on DR1-7B, Table 1). We will report intervals for matched comparisons and avoid emphasizing small competition sets without uncertainty estimates.

References

[1] Agarwal et al. On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes. ICLR 2024.

[2] Gu et al. MiniLLM: Knowledge Distillation of Large Language Models. ICLR 2024.

[3] Aggarwal and Welleck. L1: Controlling How Long a Reasoning Model Thinks with RL. COLM 2025 / arXiv:2503.04697.

[4] Luo et al. O1-Pruner: Length-Harmonizing Fine-Tuning for O1-Like Reasoning Pruning. arXiv:2501.12570, 2025.

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Official Review of Submission4012 by Reviewer B5w7

Official Review by Reviewer B5w7 📅 03 Jul 2026, 19:35 (modified: 12 Jul 2026, 23:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer B5w7

📄 Revisions (/revisions?id=jPJSlyEbs)

Paper Summary:

This paper addresses process supervision for long chain-of-thought reasoning. It argues that outcome-only rewards and conventional step-level process reward models miss the non-linear dependency structure among reasoning steps. The proposed TopoPRM extracts an implicit dependency DAG from each reasoning trace and scores both global topology quality and local continuity. The authors then build TGSD, a post-training framework combining supervised warm-start, GRPO with a correctness-first hierarchical reward, and topology-guided on-policy distillation. Experiments on several public reasoning benchmarks at 7B and 9B scales report improvements over standard GRPO while also reducing response length.

Summary Of Strengths:

1. The paper studies an interesting direction: treating long CoT as a structure with non-linear dependencies rather than a purely linear step sequence. This is relevant to process supervision and RLVR.
2. TopoPRM converts reasoning traces into dependency DAGs and combines topology quality, continuity, final correctness, and length regularization in a GRPO-based training framework.
3. The experiments cover several public mathematical and general reasoning benchmarks, and report both accuracy and token efficiency. The results suggest average gains over outcome-only GRPO in several settings.

Summary Of Weaknesses:

1. The main uncertainty is the quality of the extracted dependency DAGs. TopoPRM relies on automatically extracting graph structure from free-form reasoning traces, but the paper does not fully establish that these edges correspond to true logical dependencies. If the graph is driven by expression reuse, references, or surface continuity, it may reward formally coherent but semantically wrong reasoning.
2. The relation between topology reward and semantic correctness remains unclear. The authors acknowledge that outcome-only GRPO can outperform TopoPRM on some competition-level benchmarks at 9B. This suggests that structurally clean traces do not necessarily imply stronger key deductions. The current results may show better trace regularity and token efficiency more than reliable process supervision.
3. The full system combines SFT warm-start, hierarchical-reward GRPO, ACE, topology extraction, and on-policy distillation. Although ablations are provided, it is still hard to isolate whether the main gains come from topology-aware supervision, length control, continuity regularization, the training recipe, or distillation. Comparisons with stronger length-aware GRPO, PRM, or verifier-guided baselines would make the claim more convincing.

Comments Suggestions And Typos:

1. A small human-labeled or otherwise verifiable evaluation of extracted dependency edges would strengthen the DAG extraction claim.
2. Error cases with high topology scores but wrong answers, and correct answers with low topology scores, would clarify the limits of the reward.
3. More compute-matched comparisons with length-aware RL, process verifiers, and outcome-level reranking baselines would improve the empirical evaluation.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Limitations And Societal Impact:
The paper already notes that topology is not a proof-level verifier. The limitations would be stronger with a quantitative analysis of the structure-semantic gap and a clearer statement of where the method is expected to work or fail.

Ethical Concerns:
There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

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Response to Reviewer B5w7

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Official Comment by Authors 14 Jul 2026, 13:12 (modified: 14 Jul 2026, 22:48)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer B5w7

Revisions (/revisions?id=HOEFWLIO1i)

Comment:

We thank the reviewer. The central issue is whether the DAG is a reliable process signal or a surface-continuity artifact. R1/R2/R3 below answer W1/W2/W3; the revision narrows the claim and adds validation, failure analysis, and mechanism isolation.

W1. Surface reuse vs. support dependencies.

R1. We evaluate extracted edges against two independent references of necessary support on 120 stratified traces — a held-out Qwen3-32B annotator and human annotation — and break agreement down by edge type, the direct test of the surface-reuse concern:

Table R1. Per-edge-type precision against the necessary-support reference, with the surface-reuse vs. genuine-support interpretation of each type.

Edge type	Precision vs. reference	Interpretation
expression / claim reuse	0.59–0.63	genuine intermediate-result support
explicit order / citation	0.61	genuine step citation
variable-only overlap	0.25	surface reuse; 52% of all false positives

The only edge type that behaves like surface reuse is bare variable overlap. Overall the extractor scores $P=0.48/R=0.59/F1=0.53$ against the Qwen3-32B annotator and $P=0.58/R=0.71/F1=0.64$ against human annotation, and a second annotator (Qwen2.5-32B) agrees with the first at 0.90 raw / Cohen's kappa 0.73. Crucially, TopoPRM was built from a deployed math-grading platform (Appendix F) where professional teachers mark step-level support and errors, so the dependency notion is grounded in strict human step references from production, not only in this public-benchmark sample. We therefore

no longer describe the graph as recovering true logical dependencies; the revised claim is **surface-evidenced support dependencies** that provide useful process supervision when gated by answer correctness.

W2. Topology vs. semantic correctness.

R2. Topology complements, and is gated by, semantic correctness. We agree a topology score alone is not a correctness classifier, and our design is built precisely so that structure supplements answer correctness rather than substituting for it. Two mechanisms address the "clean structure, wrong deduction" case the reviewer raises. First, correctness gating: ACE computes advantages within the correct and wrong strata, so structural credit can only re-rank traces inside a stratum and can never lift a wrong answer above a correct one. Second, local logical-relation modeling: the continuity term scores whether each step is actually supported by its cited premises, so a globally clean but locally unsupported derivation — the dominant failure on competition problems with complex implicit structure, and a classic reward-hacking surface [6] — is penalized rather than rewarded; removing it collapses the reward (Collapse% 37.9% to 68.8%, Table 3/9), which is why continuity is a core component and not an add-on. The net effect is that topology-aware training makes structure a *stronger* predictor of correctness (the high-minus-low-topology correctness gap widens from 0.150 for outcome-only GRPO to 0.171 for full TopoPRM), i.e. the reward jointly captures answer correctness and the inter-step logical relations that outcome-only signals ignore. We add representative cases and this mechanism description to the revision.

W3. Source of the gains.

R3. We reorganize the ablation around mechanisms rather than components, on Qwen3.5-9B:

Table R2. One-at-a-time reward ablation on Qwen3.5-9B: pass@1 on GSM8K/MATH-500/AIME'24, the 8-benchmark average, and the GRPO group-collapse rate (fraction of groups with near-zero reward std).

Reward setting	GSM8K	MATH-500	AIME'24	Avg	Collapse%
Outcome-only GRPO	93.3	50.8	16.7	40.8	57.8
w/o ACE	93.8	50.8	20.0	41.7	42.1
w/o topology (length + continuity)	93.0	51.0	20.0	41.8	44.6
Full TopoPRM	94.1	51.0	30.0	45.6	37.9

Length/continuity without topology does not reach the full model, and ACE reduces reward collapse. We also ran a small best-of-N diagnostic with Qwen2.5-Math-PRM-7B [5]: TopoPRM alone is not a competitive outcome reranker, as expected, but combining PRM scores with topology slightly improves MATH-500 reranking over PRM alone (71.2 vs 70.0) in the shared pool. We present this only as evidence of complementarity, not as a replacement for PRMs or verifiers.

References

[5] Zhang et al. The Lessons of Developing Process Reward Models in Mathematical Reasoning. Findings of ACL 2025 / arXiv:2501.07301.

[6] Skalse et al. Defining and Characterizing Reward Gaming. NeurIPS 2022.

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Official Review of Submission4012 by Reviewer TsKG

Official Review by Reviewer TsKG 📅 02 Jul 2026, 23:12 (modified: 12 Jul 2026, 23:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer TsKG

📄 Revisions (/revisions?id=FyO6DjdfUo)

Paper Summary:

The paper extracts a dependency graph from a reasoning trace using a rule-plus-LLM extractor, scores its topology and continuity, and folds those scores multiplicatively into a reward that also includes outcome, format and length terms. Around this it builds a three-stage pipeline: an SFT warm start, GRPO with an asymmetric

clipping advantage estimator, and a topology-guided self-distillation step. It reports pass@5 gains over GRPO on nine benchmarks, a 15 to 24 percent token reduction, and cross-scale distillation.

Summary Of Strengths:

1. The premise is interesting: long chains of thought have non-linear dependency structure that step-local PRMs miss, and reusing one graph scorer for both reward shaping and distillation is an appealing unification.
2. The diagnostic suite is broad, covering token efficiency, collapse rate, reward-aggregation comparisons and cross-scale distillation.
3. The correctness-first multiplicative aggregation and the stratified advantage estimator are a sensible response to reward hacking in additive multi-signal rewards.

Summary Of Weaknesses:

1. The extractor at the center of the method is never validated against ground truth. All of the structural-quality gains are self-referential, and the paper itself admits the diagnostics are computed by the same extractor that drove training, so they are internal-consistency checks rather than an independent metric. There is no precision or recall of the extracted edges against human or verifier-derived support. The whole method rests on an unvalidated heuristic.
2. There are internal inconsistencies serious enough to undermine the empirical claims. The reported reasoning length is described as staying under 500 tokens in the text while the figure and a table give roughly 1568 and 1692. The main table mixes quoted and reproduced numbers and contains implausible rows, for instance an R1-distilled base scoring in the thirties on MATH and then jumping into the nineties after SFT, when that base should already be around ninety, which inflates the reported deltas. Some model names are non-standard, and one row sits far above its neighbors with no explanation.
3. The ablations look like brittleness rather than complementarity. Removing the continuity term drops the average from the mid-forties to the mid-teens, below outcome-only GRPO, with the collapse rate rising sharply. A reward whose ablation leaves the model far worse than using no process reward at all points to mis-specification or hackability, not to complementary signals.

Comments Suggestions And Typos:

- Separate the two settings clearly and state exactly what the outcome reward is on each public math benchmark. If the rubric reward was used there, the results need re-running with an answer-correctness reward.
- Explain why removing continuity drops below outcome-only GRPO. Is the topology term being hacked in that condition? A direct measurement of high-structure wrong-answer traces before and after the estimator would help.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.

Datasets: 2 = Documentary: The new datasets will be useful to study or replicate the reported research, although for other purposes they may have limited interest or limited usability. (Still a positive rating)

Software: 2 = Documentary: The new software will be useful to study or replicate the reported research, although for other purposes it may have limited interest or limited usability. (Still a positive rating)

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

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Response to Reviewer TsKG

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by Authors (Zhuoming Zhu (/profile?id=~Zhuoming_Zhu1), Xixuan Hao (/profile?id=~Xixuan_Hao2), Songxin Lei (/profile?id=~Songxin_Lei1), Yuxuan Liang (/profile?id=~Yuxuan_Liang1), +2 more (/group/edit?id=aclweb.org/ACL/ARR/2026/May/Submission4012/Authors))

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Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer TsKG

Revisions (/revisions?id=IPZPRfIhRx)

Comment:

We thank the reviewer for the reproducibility-focused review. R1/R2/R3 below answer W1/W2/W3; we separate the core validation issue from presentation/provenance errors and correct both.

W1. Self-referential extractor diagnostics.

R1. The submitted structural diagnostics were internal-consistency checks computed by the same extractor that drove training. We now add external edge-level validation on 120 stratified traces, labeled for pairwise necessary support under two references computed entirely separately from the training pipeline: a held-out Qwen3-32B annotator and human annotation. Against the Qwen3-32B annotator the extractor scores $P=0.48/R=0.59/F1=0.53$, and against human annotation $P=0.58/R=0.71/F1=0.64$; a second architecturally distinct annotator (Qwen2.5-32B) reaches 0.90 raw agreement / Cohen's kappa 0.73 with the first, confirming the reference is not one model's artifact. The method additionally originates from a deployed math-grading platform (Appendix F) where professional teachers provide strict step-level support references; we present this as motivation for the dependency notion, not as a substitute for the new edge-level validation above. We will release the validation protocol and labeled edge set, and rescope the method as a topology-aware process signal rather than a proof verifier.

W2. Length and Table 1 inconsistencies.

R2. The "<500 tokens" phrase came from a training-dynamics plot and should not have been stated as an evaluation-time length; we remove it and report benchmark-specific evaluation lengths from the same script used for accuracy. The correct wording is "up to 24% fewer tokens on GSM8K, about 13% on the four-primary-benchmark mean," not a universal 15–24% reduction. We also split Table 1 into quoted reference rows and reproduced matched rows, remove bold/underline comparisons across non-matched settings, and add checkpoint IDs, decoding settings, and metric definitions per row. The flagged DeepSeek-R1-Distill-Qwen-7B MATH row (36.8 vs later 92.8) mixes decoding protocol and provenance in a way that can be misread, so rather than defend a specific interpretation we move such ambiguous-provenance rows to a reference-only appendix and compute deltas only within audited matched blocks (same base, SFT, decoding, and metric).

W3. w/o-continuity collapse.

R3. This ablation reveals hackability of topology alone. Without continuity, global checks such as acyclicity and no-orphan conclusions can be satisfied by sparse, formulaic traces that skip local support; the reward then collapses and the model falls below outcome-only GRPO. The interpretation is not "all structural components independently help," but that topology, local continuity, and correctness-stratified ACE are jointly necessary.

Table R1. Effect of removing continuity vs. ACE on Qwen3.5-9B: 8-benchmark average pass@1 and the GRPO group-collapse rate.

Setting	Avg pass@1	Collapse%
Outcome-only GRPO	40.8	57.8

Setting	Avg pass@1	Collapse%
w/o continuity	16.3	68.8
w/o ACE	41.7	42.1
Full TopoPRM	45.6	37.9

Outcome reward clarification. Public-benchmark rewards are exact final-answer rewards: boxed-answer extraction with benchmark-specific normalization/equivalence for GSM8K, MATH-500, OlympiadBench, Omni-MATH, AIME, and CNMO; option correctness for MMLU/GPQA-Diamond. The rubric-score reward is used only for the in-domain critique appendix. We separate these definitions in Appendix D with pseudocode, so the public-benchmark results do not require rerunning for this issue.

References

[7] DeepSeek-AI. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning. arXiv:2501.12948, 2025.

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